

SAP Inventory Management

Automate Purchase Order Generation

Identify actors and goals

Actors and Goals:

SAP System (Automated Actor)

- Monitor inventory levels of raw materials in real-time.
- Automatically generate purchase orders when stock hits the reorder point.

Warehouse Manager

- Define and adjust reorder points and ideal stock levels for materials like X812 or J223.
- Ensure production continuity by preventing stockouts.

Define preconditions and postconditions

Selected Goal: SAP System automatically generates a Purchase Order when stock levels fall below the threshold.

Preconditions:

- The raw material (e.g., metals, plastics) is correctly defined in the SAP Material Master.
- A specific reorder point and ideal stock level have been configured for each Material ID.
- Valid vendor information and lead times are maintained in the system.

Postconditions:

- A formal Purchase Order (PO) is generated and sent to the appropriate supplier.
- The system inventory status is updated to reflect that a replenishment order is in progress.

Define the steps

Steps:

- **Trigger:** The inventory level for a material (e.g., Material S549) falls below its defined reorder point.
- **Step 1:** The SAP system detects the stock shortage through real-time monitoring.
- **Step 2:** The system calculates the required quantity by comparing the current stock against the "Ideal Stock" level.
- **Step 3:** The system identifies the primary vendor and checks the associated lead time for that material.
- **Step 4:** The system automatically generates a Purchase Order document with the calculated quantity and vendor details.
- **Step 5:** An automated notification is sent to the Purchasing Agent for awareness.

Alternative Flows:

- **Manual Override:** If a primary supplier is unavailable or a price change is detected, the system flags the PO for manual review by the Purchasing Agent instead of automatic dispatch.
- **Incomplete Data:** If lead time or vendor data is missing, the system generates an error log and alerts the Warehouse Manager.

Additional Notes and Assumptions

- It is assumed that the SAP Material Management (MM) module is fully integrated with current warehouse data.
- Manual intervention is still permitted for exceptional circumstances to maintain flexibility.
- The system must prioritize materials marked as "Stockout" (e.g., X812, S549) to minimize production delays.